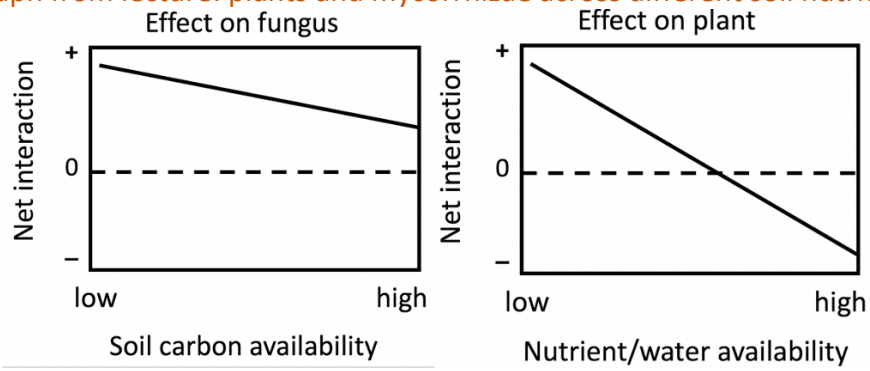
Study Guide for Ecosystems KEY

1. Photosynthesis and Dissolution
2. Blank1: the amount of solar radiation; blank2: the amount of available nutrients
3. A
4. E
5. A
6. While it seems counter-intuitive, positive relationships can be more common under stressful conditions because each partner is more likely to benefit from the relationship. Under non-stressful conditions, individuals can obtain the nutrients they need and perform life functions like reproduction without 'help' from other species, but when conditions are stressful or more difficult, they are more likely to benefit from cooperation.
   Example graph from lecture: plants and mycorrhizae across different soil nutrient levels



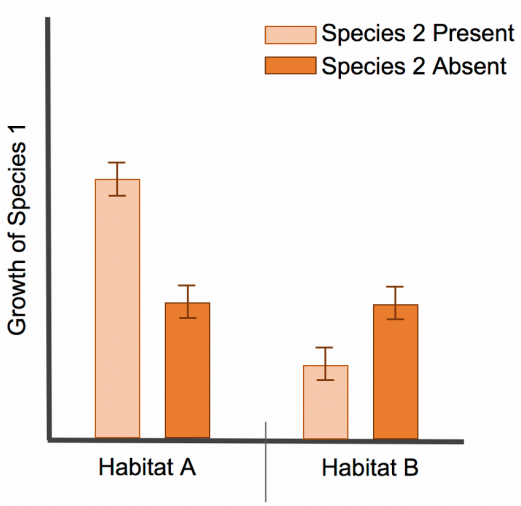
7. The clearest example of this is the plant/mycorrhizae relationship. Under low soil nutrient conditions, both the mycorrhizae and the plant benefit: the plant is able to access soil nutrients through the fungal hyphae network (not just its own roots) and the mycorrhizae gets sugar from the plant's photosynthesis. Under high soil nutrient conditions, the mycorrhizae still benefits (though possibly less strongly) since a non-photosynthetic organism will pretty much always benefit from getting sugar from a photosynthetic organism. The plant however, may be negatively impacted if soil nutrient levels are high enough that the plant could access all the nutrients it needs through its own root network and does not need the fungal hyphae network. There may even be a small region of moderate soil nutrient availability when this relationship is commensal (the mycorrhizae benefit and the plant is not strongly impacted either way.

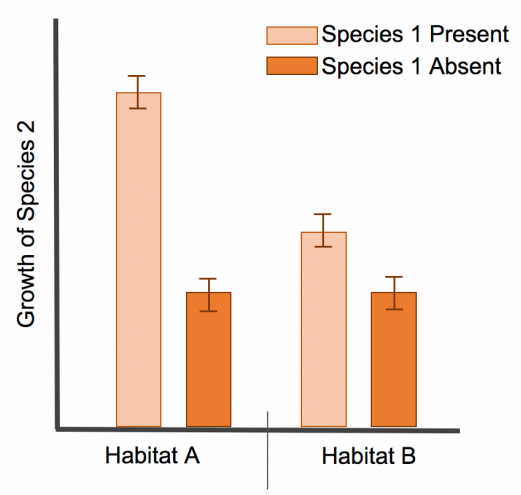
8. Primary succession is succession that begins on bare ground with no soil profile. Examples from lecture include: glacier retreat, dune accretion, lava flows.
   Secondary succession is succession that has a developed soil foundation (even if everything above ground has been destroyed, though in some cases, some above-ground organisms can survive). Examples from lecture include: fire, land use change (like abandoned agricultural land)

9. Generally speaking, we expect pioneer species to be r-selected and climax community species to be primarily K-selected (especially if we are focusing on the plant community). This is not to say that no climax community species will be r-selected, just that we expect the majority of early pioneer species to be r-selected and most of the K-selected species to represent the climax community

10.

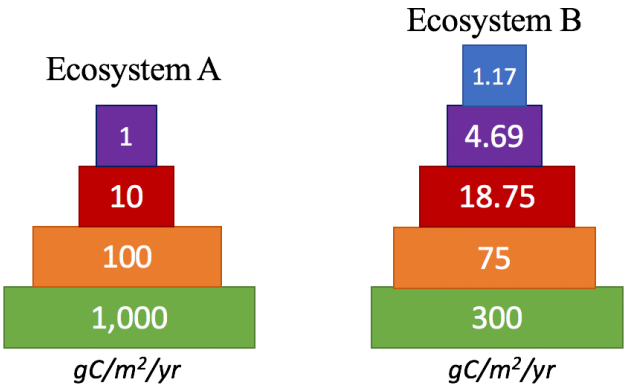
Conditional: the direction of impact (positive or negative) of the presence of one species on the other is different in Habitat A than in Habitat B. The impact on the other species may or may not vary by habitat. This example represents the relationship between plants (Species 1) and mycorrhizae (Species 2) in low nutrient (A) and high nutrient (B) ecosystems.





11.

a. Ecosystem B
b. Ecosystem A, because it has more energy available at the primary consumer level and endotherms generally require more energy than ectotherms
c. 4 trophic levels for Ecosystem A, 5 for Ecosystem B
d. Ecosystem A starts with >3x the primary productivity as Ecosystem B; however, Ecosystem B has 2.5x the efficiency of Ecosystem A, which means it has more energy available at the upper trophic levels, and indeed can support an entire additional trophic level than Ecosystem A. In this way, efficiency of energy transfer has a larger impact on the energy availability at higher trophic levels.



12. Each of the efficiency terms focus on a particular 'step' of the process of energy movement between trophic levels. Consumption efficiency relates to how much of the biomass available at the lower trophic level is consumed by the higher trophic level. Assimilation efficiency focuses on how much of the biomass that is consumed is digested (and not excreted). Production efficiency focuses on how much of the biomass that is digested is converted into biomass at the higher trophic level (and not burned for metabolic processes like respiration). All of these together represent ecological efficiency, which is the amount of total biomass at the higher trophic level compared to the lower.

13.

a. For this question, many students choose the herbivore; however, digesting the cell walls in plant cellulose is actually extremely difficult for most organisms. You may already be aware of some digestive adaptations that some herbivores have for dealing with this problem, such as multiple stomachs, chewing cud, fermentation chambers in their gut, etc. Some animals, like rabbits, even perform 'coprophagy' where they eat their feces to digest it a second time. Consequently, carnivores generally have higher assimilation efficiency than herbivores.
b. The loss pathway in production efficiency is primarily respiration. Endotherms have much higher respiration rates than ectotherms in order to maintain body temperatures. Consequently, we would expect ectotherms to have higher production efficiency because the loss pathway is minimized.
c. The plant community in a grassland is mostly grasses, while in a forest it is mostly trees. Most organisms cannot eat wood, so the amount of plant biomass that they can consume is much smaller in a forest than a grassland. Consequently, we would expect consumption efficiency to be higher in grassland herbivores than in forest herbivores.
d. The C3 plant would likely have higher production efficiency because it is more energy efficient. C4 plants are more water efficient, but spend more energy building sugar (because of moving CO_2 into the bundle sheath cells against the concentration gradient)

14. Terrestrial net primary productivity is driven primarily by temperature (and influenced by precipitation when it is extreme). So, the tropics have high productivity because they are hot and wet. Marine net primary productivity is driven by nutrient availability, not temperature or precipitation. Coastal areas have the highest nutrients due to runoff of fertilizer and waste products from land, and because continents drive upwelling of deep ocean currents (which contain lots of nutrients) to the surface.

15. Eating at a lower trophic level is more energy efficient from an ecosystem level perspective (though your assimilation efficiency may be reduced due to the cellulose issue described in Q3 above). You can think of it in two ways, either through the efficiency steps, or in terms of the energy pyramids. For example, you have to feed a cow a lot of corn in order for it to produce meat. Each of those efficiency steps (especially assimilation and production) lead to there being less meat mass on the cow than there was corn mass fed to it. If you just ate the corn mass directly, you bypass those inefficiencies. Second, in terms of the energy pyramids, moving to a lower trophic level increases the amount of energy available to you at the trophic level you consume because lower trophic levels always contain more energy than higher levels.

16. Dead zones occur when high nutrient inputs produce an algal bloom that then dies and decays, which depletes the oxygen in lower water levels to the point that many organisms suffocate. They are commonly found at the mouths of large rivers because watersheds concentrate terrestrial runoff into rivers, which produces a large input of nutrients in a small area of the ocean. The Mississippi River produces a dead zone each year in the Gulf of Mexico due to its enormous watershed (3.2 million square kilometers; 40% of the continental US) and the prevalence of agricultural land and municipal waste within the watershed.

17. Groundwater is important for agricultural irrigation, especially in the Midwest United States, where 20% of our industries of corn, wheat, cotton, and beef all rely on the Ogallala Aquifer. The fact that groundwater is 'disconnected' from the rest of the water cycle, however, means that it recharges extremely slowly, much more slowly than we withdraw water. Consequently, this source of water is increasingly difficult and expensive to access and may no longer even be accessible at any cost by 2150.

18. Energy diagrams must always take the shape of a pyramid (each higher level smaller than the level below it) because the primary producers (bottom of the pyramid) are the only step at which energy enters the system (through photosynthesis). At all other trophic levels, energy is lost through respiration and other forms of inefficiency, and no other energy can enter the system, producing a pyramid shape.

19. Removing filter feeders will reduce the herbivory on algae and likely cause an increase in algal growth, reducing water clarity. Introducing Northern Pike will cause a trophic cascade of increased predation pressure on planktivorous fish, reducing their population and causing decreased predation pressure on zooplankton, increasing their population and increasing herbivory pressure on algae, maintaining or even increasing the water clarity. If all you consider is water clarity, then the Northern Pike might be the best option. You should, however, strongly caution the town about the dangers of introduced species and possible unintended consequences of the presence of Northern Pike beyond water clarity.

20. A top-down trophic cascade is one in which the initial change in the system occurs near the top of the trophic levels and the impacts of the change cascade down to lower trophic levels. The example in lecture was the orca → sea otter → urchin → kelp example. A bottom-up trophic cascade is one in which the initial change in the system occurs at the bottom of the trophic levels and the impacts of the change cascade up to higher trophic levels. The example in lecture was the nutrients for aquatic plants, which led to algal blooms and fish death.

21. Aquifers are similar to fossil fuels in that they come out of the ground, and are very slow to replenish (so, essentially non-renewable). Aquifers are different though in that they did not form from fossil organisms (as coal and oil did).

22.
    a. The largest pool is the atmosphere (shown in the tan box at the top of the diagram). The largest flux is the unlabeled flux that connects the pool of marine biomass and dissolved nitrogen in the ocean (this flux is unlabeled because it represents multiple processes, the relative contributions of which are not well understood)

    b. Livestock/agriculture, biomass burning, fossil fuel burning, industrial N fixation, and agricultural N fixation. Together, these contribute 81 units into the atmospheric pool (compared to 268 from the natural denitrification fluxes), and 130 into the plant biomass pool (compared to 128 from natural N fixation), so human inputs into the plant biomass pool exceed natural inputs.

c. One possible pathway: Starting in the atmosphere, it could move through atmospheric deposition onto the landscape, then move through runoff into the pool of dissolved nitrogen in the ocean, then through denitrification back to the atmosphere.